SWITCHING BASICS

Theory and Cisco Packet Tracer Practical Report

Objective

To understand how an Ethernet switch learns MAC addresses, builds its MAC address table, and forwards frames. The practical uses one switch and three PCs, followed by a frame-forwarding scenario.

Theory: How a Switch Works

A network switch connects devices on a local area network (LAN). When an Ethernet frame arrives, the switch reads the source and destination MAC addresses. It learns the source MAC address by saving the address together with the port where the frame arrived. This information is stored in the MAC address table, also called the forwarding table.

If the destination MAC address is already in the table, the switch sends the frame out only through the matching port, provided that port is not the same port on which the frame arrived. If the destination is unknown, the switch floods the frame out the other eligible ports in the same VLAN. Broadcast frames are also flooded within the VLAN. The switch does not send a frame back out the port on which it arrived.

Important Terms

- **MAC address:** A link-layer address used to identify a network interface on an Ethernet LAN.
- **MAC address table:** A table mapping learned MAC addresses to switch ports and VLANs.
- **Learning:** The switch records the source MAC address and the incoming port from each received frame.
- **Known unicast:** A frame whose destination MAC address is already present in the table; it is forwarded to the matching port.
- **Unknown unicast:** A frame whose destination MAC is not in the table; it is flooded to eligible ports in the same VLAN.
- **Aging:** Learned dynamic entries are removed after a period without traffic, allowing the table to stay current.

Practical Setup

Create a simple star network using one Cisco switch and three PCs.

Device	Switch port	Example IP address	Subnet mask
PC1	FastEthernet0/1	192.168.1.10	255.255.255.0
PC2	FastEthernet0/2	192.168.1.20	255.255.255.0
PC3	FastEthernet0/3	192.168.1.30	255.255.255.0

Use copper straight-through cables from each PC to the switch. Set the IP addresses on each PC using Desktop > IP Configuration. A default gateway is not needed for communication within this same subnet.

Task 1: Check the MAC Address Table Before Communication

1. Open Packet Tracer and place one switch (for example, a 2960) and two PCs.
2. Connect PC1 and PC2 to separate switch ports and configure their IP addresses.
3. On the switch CLI, enter privileged EXEC mode and run the command below.

```
Switch> enable
Switch# show mac address-table
```

At first, the dynamic MAC address table may be empty, or it may contain entries created by earlier traffic. If it contains entries, clear dynamic entries if supported by the device, or use a fresh switch and inspect the table before sending test traffic. Link-up or control traffic can affect what appears.

Task 2: Ping PC1 to PC2

4. From PC1, open Desktop > Command Prompt and enter: ping 192.168.1.20
5. Run show mac address-table on the switch again.
6. Record the learned MAC addresses and the ports where they were learned.
7. Ping in the reverse direction from PC2 to PC1 and check the table again.

A ping typically triggers ARP first if the sender does not already know the other PC's MAC address. The switch learns the source MAC from frames it receives. After the ARP exchange and ICMP traffic, entries for both PCs should normally be visible. The exact timing depends on the simulator and existing caches.

Task 3: Add PC3

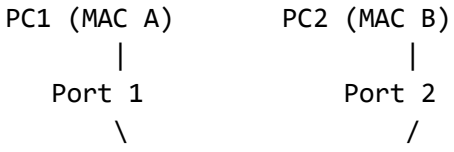
8. Connect PC3 to another switch port and assign 192.168.1.30/24.
9. From PC3, ping PC1 using: ping 192.168.1.10
10. Run show mac address-table and record the entries for PC1, PC2, and PC3.
11. Confirm that each learned MAC address is associated with the port where that device is connected.

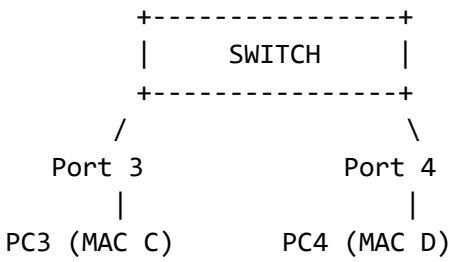
Task 4: Explain a Specific Frame Forwarding Example

Given table: Port 1 — 00:11:22:33:44:55; Port 2 — 66:77:88:99:AA:BB. A frame arrives on Port 1 with destination MAC 66:77:88:99:AA:BB.

12. The switch receives the frame on Port 1 and reads its source and destination MAC addresses.
13. It learns or refreshes the source MAC address on Port 1, associating that source with the incoming port.
14. It searches the MAC address table for destination 66:77:88:99:AA:BB and finds it on Port 2.
15. Because the destination is on Port 2, the switch forwards the frame only out Port 2, not out every port.
16. The destination device receives the frame and processes it if the frame is valid and addressed to it.

Task 5: Four-Device Learning and Forwarding Diagram





Example in my own words: When PC1 sends a frame, the switch notices PC1's source MAC address and remembers that it is connected to Port 1. If the destination MAC is already known, the switch sends the frame through the port recorded for that address. If the destination is not known yet, the switch sends the frame through the other ports in the same VLAN so the destination can receive it. As PCs communicate, the switch learns more addresses and can forward later frames more directly.

Observation Table

Test stage	Command / action	Expected observation	Actual result
Before traffic	show mac address-table	No dynamic PC entries yet, if table is fresh.	_____
PC1 pings PC2	ping 192.168.1.20	Switch learns source MACs from traffic; both PC entries normally appear after ARP/ping.	_____
PC3 pings PC1	ping 192.168.1.10	PC3's MAC is learned on its port; existing entries remain while active.	_____
Known destination	Inspect table and send frame	Frame is forwarded only to the destination's mapped port.	_____

Useful Switch Commands

- **show mac address-table —** Displays learned MAC addresses, VLANs, and switch ports.
- **show mac address-table dynamic —** Displays dynamically learned entries, if supported.
- **show interfaces status —** Checks switch port status on supported models.
- **clear mac address-table dynamic —** Clears dynamic MAC entries on many Cisco switch models; availability may vary.

Conclusion

A switch learns MAC addresses from the source address of incoming frames and associates each address with the port where it was seen. It forwards known unicast frames to the correct port and floods unknown unicast or broadcast frames to eligible ports within the same VLAN. The Packet Tracer activity demonstrates how the MAC address table grows as PCs communicate. Fill in the observation table with the results seen in your lab.